

Excel-Based Product Inventory Report

Dataset Description:

This dataset contains information about various products from different countries, including their manufacturing details, brand information, pricing, inventory quantity, and product categories. It is designed for practicing data cleaning, data transformation, conditional formatting, and inventory analysis in MS Excel.

Column Description:

Column Name	Description
Manufacturing Date	Date when the product was manufactured.
Country Code	Country identifier where the product is associated.
Product ID	Unique identifier for each product.
Product Brand	Product brand or model information.
Product Name	Name of the product.
Brand Name	Manufacturer or brand of the product.
Price (\$)	Selling price of the product.
Quantity	Available inventory quantity.
Category	Product classification (Electronics, Fashion, Kitchen, Outdoor, Accessories, etc.).

Data cleaning & transformation:

1. Handling Missing Values:	
A. Check for missing values in the 'Price' column. How would you handle products with missing price information?	
Formula a. =AVERAGE((D2:D35))	b. =IF(ISBLANK(D2),AVERAGE(\$D\$2:\$D\$35),D2)
Average Value	309.35
Ans: Missing prices will be replaced with the average price.Existing prices remain unchanged.	
B. If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.	
Formula =IF(ISBLANK(F2),"Unknown",F2)	
Ans: Missing categories were replaced with "Unknown", while the existing price values remained unchanged.	

2. Correcting Inconsistent Data:

A. Identify any inconsistent text formats present in the "Product Name" column.

Formula =PROPER(TRIM(CLEAN(B2)))

Ans: The "Product Name" column was cleaned by using the PROPER, TRIM, and CLEAN functions. This removed extra spaces, unwanted characters, and corrected text formatting to maintain consistency across all product names.

B. Identify any typos present in the "Category" column.

Ans: Yes, there is one typo error in Category column

C. Use the find and replace function to standardize the text formats in the "Product Name" column and fix any

Steps to find and replace the typos

1. Select the Category column
2. Go to Home tab
3. Go to Editing column, Select Find & Select
4. Select Replace command
5. In the **Find what** box, type the incorrect category name (typo).
6. In the **Replace with** box, type the correct category name.
7. Click **Replace** to correct one entry or **Replace All** to correct all matching entries.
8. Repeat the process for other typos in the Category column.

Ans: All typo errors in the Category column were identified and corrected to maintain consistent category names.

3. Removing Duplicates:

A. Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.

Steps to Remove Duplicate Rows

1. Select the entire dataset or table.
2. Go to the Data tab on the Ribbon.
3. Click Remove Duplicates in the Data Tools group.
4. A dialog box will appear.
5. Select the columns you want to check for duplicates.
6. Click OK.
7. It will remove the duplicate rows and display the number of duplicates removed and unique values remaining.

Ans: In this dataset, 3 duplicate rows were found and removed successfully.

4. Splitting and Merging Data:

A. Split the "Product ID" column into two separate columns for "Manufacturing Date" and "Country Code". Remove unnecessary characters, if any.

a. Formula =TRIM(CLEAN(A2)) b. Formula =LEFT((A2),6) c. Formula =RIGHT((A2),2)

Ans: Unnecessary spaces and unwanted characters were removed using the TRIM and CLEAN functions. The "Product ID" column was split into two separate columns: "Manufacturing Date" and "Country Code" using LEFT and RIGHT formulas.

B. Merge the "Brand Name" and "Product Name" columns into one column named "Product Brand".

Formula =CONCATENATE(B2," - ",C2)

Ans: The "Brand Name" and "Product Name" columns were merged into a new column named "Product Brand" using the CONCATENATE operator to create a combined product description.

5. Number Formatting:

A. Format the data type of the "Price" column to a currency format.

Steps to change the data type of price column

1. Select the Price column.
2. Go to the Home tab.
3. In the Number group, click the dropdown menu.
4. Select Currency or Accounting format.
5. Choose the required currency symbol (₹, \$, €, etc.).
6. Adjust decimal places if needed.
7. Click OK or apply the format.

Ans: The data type of the "Price" column was changed from text format to currency format using the Format Cells option.

B. Format the "Manufacturing Date" column to display dates in the "DD-MM-YYYY " format.	
Steps to Format the "Manufacturing Date" Column into DD-MM-YYYY Format	
1. Select the Manufacturing Date column.	
2. Go to the Home tab.	
3. In the Number section, click the dropdown menu (currently may show General or another format).	
4. Choose More Number Formats (at the bottom of the list).	
5. In the Format Cells dialog, go to the Number tab.	
6. Select Date (or Custom if you want to manually type the format).	
7. Under Type, choose or type DD-MM-YYYY.	
8. Click OK.	
Ans: The Manufacturing Date column is now formatted to DD-MM format. Since the dataset does not contain year values, all dates are displayed in a consistent and easy-to-read day-month format..	

6. Conditional Formatting:	
A. Apply data bar or color scales conditional formatting in the "Price" column.	
Steps to change the colour of the price column using conditional formatting	
1. Select the Price column.	
2. Go to the Home tab.	
3. In the Styles section, Click Conditional Formatting.	
4. Move the cursor near the Data Bar, a side menu pops out automatically	
5. From that side menu, select a format that suits the current column (e.g., Blue Gradient Fill or Orange Solid Fill).	
Ans: The Data Bar conditional formatting was successfully applied to the Price column. This makes it easier to compare product prices	

B. Create a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."	
Steps to create a custom rule for conditional formatting in the "Category" column	
1. Select the Category column.	
2. Go to the Home tab.	
3. In the Styles section, Click Conditional Formatting.	
4. Move the cursor near the highlight cell rule, a side menu pops out automatically	
5. From that side menu, select Text that contain.	
6. In the dialog box, type: Electronics	
7. Choose a formatting style or custom color to highlight the cells.	
8. Click OK.	
Ans: A conditional formatting rule was created in the "Category" column to highlight all cells containing the category "Electronics," making them easier to identify and analyze.	

Conclusion:

This project successfully demonstrated the process of cleaning, formatting, and analyzing a product inventory dataset using Microsoft Excel. The dataset was reviewed for missing values, text inconsistencies, and formatting issues.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Product ID2	Manufacturing Date	Country Code	Product Brand
28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	28-JAN-US	28-JAN	US	Laptop - Dell
15-FEB-US	Sneakers	Nike	\$80	15	Fashion	15-FEB-US	15-FEB	US	Sneakers - Nike
03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	03-MAR-US	03-MAR	US	Coffee Maker - Keurig
11-APR-US	Smartphone	Samsung	\$900	25	Electronics	11-APR-US	11-APR	US	Smartphone - Samsung
22-MAY-US	Backpack	North Face	\$70	20	Unknown	22-MAY-US	22-MAY	US	Backpack - North Face
07-JUN-UK	Headphones	Sony	\$309	45	Electronics	07-JUN-UK	07-JUN	UK	Headphones - Sony
19-JUL-UK	T-Shirt	Adidas	\$30	5	Fashion	19-JUL-UK	19-JUL	UK	T-Shirt - Adidas
23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	23-AUG-UK	23-AUG	UK	Blender - Ninja
05-SEP-UK	Tablet	Apple	\$500	50	Electronics	05-SEP-UK	05-SEP	UK	Tablet - Apple
14-OCT-UK	Hiking Boots	Timberland	\$130	10	Outdoor	14-OCT-UK	14-OCT	UK	Hiking Boots - Timberland
17-JUN-IN	Laptop	HP	\$950	25	Electronics	17-JUN-IN	17-JUN	IN	Laptop - HP
25-NOV-AU	Sneakers	Adidas	\$90	40	Unknown	25-NOV-AU	25-NOV	AU	Sneakers - Adidas
08-DEC-DE	Coffee Maker	Nespresso	\$120	35	Unknown	08-DEC-DE	08-DEC	DE	Coffee Maker - Nespresso
18-FEB-CA	Smartwatch	Fitbit	\$150	15	Electronics	18-FEB-CA	18-FEB	CA	Smartwatch - Fitbit
16-APR-ES	Headphones	Bose	\$250	20	Electronics	16-APR-ES	16-APR	ES	Headphones - Bose
21-AUG-CA	Laptop Bag	Samsonite	\$50	35	Accessories	21-AUG-CA	21-AUG	CA	Laptop Bag - Samsonite
20-AUG-CN	Smartwatch	Huawei	\$160	15	Electronics	20-AUG-CN	20-AUG	CN	Smartwatch - Huawei
27-JAN-IT	Laptop	Asus	\$980	10	Electronics	27-JAN-IT	27-JAN	IT	Laptop - Asus
01-MAR-UK	Sunglasses	Oakley	\$150	15	Fashion	01-MAR-UK	01-MAR	UK	Sunglasses - Oakley
14-AUG-US	Camping Tent	Coleman	\$309	10	Outdoor	14-AUG-US	14-AUG	US	Camping Tent - Coleman
14-MAY-RU	Camera	Nikon	\$700	50	Electronics	14-MAY-RU	14-MAY	RU	Camera - Nikon
09-JAN-CA	Microwave	Panasonic	\$80	20	Kitchen	09-JAN-CA	09-JAN	CA	Microwave - Panasonic
19-JUL-BR	Fitness Tracker	Xiaomi	\$150	30	Unknown	19-JUL-BR	19-JUL	BR	Fitness Tracker - Xiaomi
29-SEP-CA	Smartphone	Google	\$800	45	Electronics	29-SEP-CA	29-SEP	CA	Smartphone - Google
03-JUN-CA	Sunglasses	Ray-Ban	\$309	25	Fashion	03-JUN-CA	03-JUN	CA	Sunglasses - Ray-Ban
11-JUL-CA	Blender	Vitamix	\$400	40	Kitchen	11-JUL-CA	11-JUL	CA	Blender - Vitamix
07-MAR-CA	Dress	Zara	\$60	30	Fashion	07-MAR-CA	07-MAR	CA	Dress - Zara
13-APR-CA	Toaster	Hamilton	\$40	10	Kitchen	13-APR-CA	13-APR	CA	Toaster - Hamilton
24-MAY-CA	Fitness Tracker	Garmin	\$130	5	Electronics	24-MAY-CA	24-MAY	CA	Fitness Tracker - Garmin
02-DEC-CA	Jeans	Levi's	\$50	50	Fashion	02-DEC-CA	02-DEC	CA	Jeans - Levi's
09-JUL-FR	Watch	Casio	\$100	20	Accessories	09-JUL-FR	09-JUL	FR	Watch - Casio